ARIHO JUSTUS

B35874

S25M19/043

Python in action – Exploring Object – Oriented Programming in the Real World

Abstract

This survey paper provides a review of Python programming in real world. It looks at places which commonly use object oriented programming. The advantages and disadvantages of Python are compared with other programming languages.

The results indicate that the simplicity, strong libraries and community support of Python has made it popular.

Despite the fact that, limitations such as slow execution speed, high memory consumption and others. In the end, the paper discusses areas to improve and research further.

Introduction

Python, object oriented programming language is widely used because it is easy to learn, teach, understand, use, install and deploy. In addition, python has a large ecosystem of libraries.

It uses the four pillars that is; encapsulation, inheritance, polymorphism and abstraction to make it widely used in different fields.

A study is being conducted into applications of python in object-oriented programming as it addresses daily life problems. Moreover, as most of the real world problems are object classified, this seems like an important project.

Even though it is widely used, there are limitations that need to be fixed. These limitations will improve python and tackle more real world problems.

Survey information.

The survey highlights Python's four important pillars of object-oriented programming which are used while programming in Python to solve the problems of the real world.

The four pillars of OOP are abstraction, inheritance, encapsulation and polymorphism.

The four pillars together provide structured approaches to build reusable software applications using Python.

Python is a programming language making waves in the software and web development industry, primarily because it is object-oriented. The object-oriented

nature of C++ makes it easy for developers to structure code into reusable code and manageable components, which is highly advantageous when creating large and complex systems.

Several big companies use Python applications and these include; Instagram, Google, YouTube, Dropbox, Reddit, Pinterest, Spotify and Netflix. These organizations use Python for backend development, data processing, task automation, and machine learning and so on. Python is popular amongst them, which showcases its versatility, as it can be used for small-scale applications and large software.

Object oriented programming with Python is professionally utilized in building web applications, financial and bank systems, healthcare systems, games, automation software, data/Artificial intelligence applications, desktop applications, APIs and backend services.

Conclusion/ Future works.

Despite being a versatile and powerful programming language, Python has a number of limitations. One of the major challenges of Python is the slow execution speed, compared to a compiled language like C, C++, and Rust. With applications requiring a lot of computational processing, this becomes noticeable.

Another constraint is high memory utilization. The reason Python programs use more memory is because of the objects and the dynamic typing used by Python. Although it's possible to use Python to create mobile applications, it is usually not the main language for native mobile app development.

Even web browsers have some limitations in python programming. In order to execute a Python program, it should be first compiled. Even though Pyodide-type technology lets one run Python in the browser, the use cases aren’t as popular as JavaScript.

Another problem is the Global Interpreter Lock (GIL) in typical CPython installations. The GIL prevents multiple threads from executing CPU-bound python code at the same time. Recent versions of the free-threaded python address this limitation even if they are still in use and compatible.

Because Python is a dynamically typed programming language, some errors related to typing are detected only at run time and not at compile time. This may make some bugs hard to detect in large apps. Packaging and deploying an application can be quite difficult, especially with dependencies and Python environments.

Lastly, Python is not the most suitable option for low-level programming. In comparison to C, Rust, and other strict programming languages, Python enables lower access to memory and hardware.

Despite the numerous restrictions, Python is easy to use, has numerous libraries, is readable and has a wide use.

Reference

S. F. Lott and D. Phillips, *Python Object-Oriented Programming*, 4th ed. Birmingham, U.K.: Packt Publishing, 2021.

Python Software Foundation, “Python documentation,” *Python Documentation*. [Online]. Available: Python Documentation

Python Software Foundation, “Free-threading HOWTO,” *Python Documentation*. [Online]. Available: Free-threading Python documentation